

Lab 5, Act 2: When Calculus Runs Out

Does highway noise quiet the frog chorus?

NRES 746, name: _____ group: _____

The scenario

Males of the frog species from Act 1 call to attract mates, and traffic noise can drown them out. A highway runs through the spring complex. To see how far the noise reaches, you visit 15 ponds at different distances from the road and count the calling males you hear during one 5-minute listen at each pond. Here are your observed measurements:

Pond	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Distance (km)	0.1	0.2	0.3	0.5	0.7	1	1.2	1.5	2	2.5	3	4	5	6	8
Males heard	1	0	1	2	6	4	8	11	13	5	12	6	11	12	14

A reasonable model is that the expected count rises with distance and levels off far from the road:

$$\lambda(d) = a(1 - e^{-bd})$$

This is the monomolecular function from Bolker's bestiary (Ch. 3). Bolker's worked examples in Ch. 6 (p. 243-249) are a useful guide for this Act (but this problem also differs in several ways).

How far from the road does the chorus recover, and how strong is the evidence that the road matters at all?

Try Part 1 on paper first, before you open a laptop or start typing code.

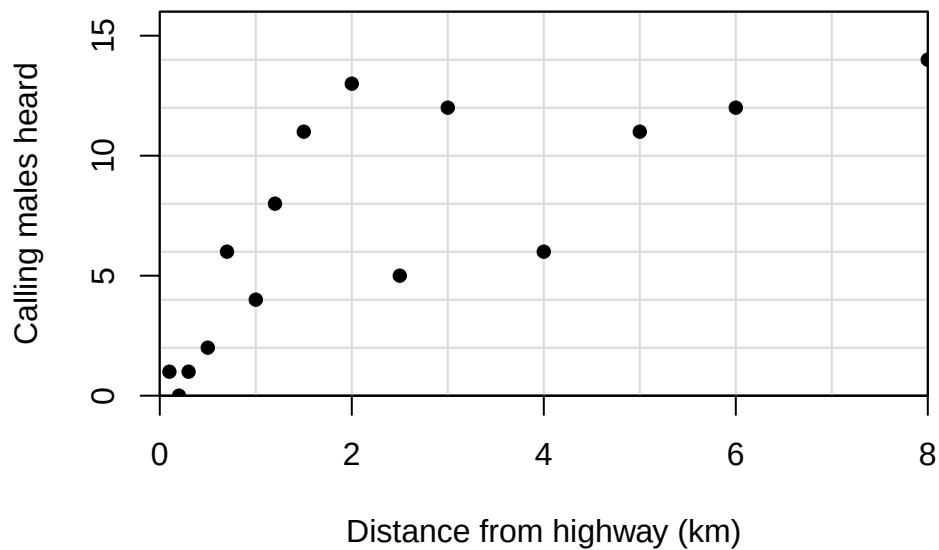
Part 1: Plan it on paper (individual, ~25 min)

1a. What do a and b mean biologically, and what are their units?

1b. Act 1 used a binomial distribution. Why is a Poisson distribution a more reasonable choice for these counts?

1c. Use the figure below and your function to estimate starting values for a and b . Show how you got your estimated starting value for b .

Hint for b : from the figure, estimate the distance $d_{1/2}$ at which the chorus is halfway to its maximum, so that $\lambda(d_{1/2}) = a/2$. Then solve that equation for b . Bolker's bestiary of functions (Ch. 3) shows how parameters like these relate to features of the curve.



$a \approx$ _____ $b \approx$ _____

1d. Write the negative log-likelihood, $NLL(a, b)$, for the 15 ponds using the Poisson pmf. Simplify it as far as you can. As in Act 1 (1c), you can drop any term that doesn't depend on a or b , since it won't change where the minimum is.

In Act 1, calculus gave a formula for $\hat{p}$. That rarely happens with real models: for this one, no algebra trick produces a formula for $\hat{a}$ and $\hat{b}$. In practice we usually let a computer search for the minimum numerically instead, which is what you'll plan next.

1e. In plain English (not R code), describe step by step what a function must do to compute the NLL for one candidate pair (a, b) . Then describe how you would use that function to find the MLE.

Checkpoint: finalize your plan (~5 min)

Look back over 1c-1e and fix anything that no longer looks right. Then choose one path for Part 2:

- **Path A (AI-assisted).** Write a prompt, based on your plan from 1e, asking an AI assistant for R code. Record your prompt (not the code it returns) and your results.
- **Path B (self-coded).** Write the R code yourself, without AI. R's help pages, web search, and classmates are fine.

Path chosen: _____

Part 2: Fit the model (laptop, ~50 min)

The data are in `FrogChorus.csv` (columns `pond`, `dist_km`, `males`), or you can paste them into R directly:

```
dist_km <- c(0.1, 0.2, 0.3, 0.5, 0.7, 1, 1.2, 1.5, 2, 2.5, 3, 4, 5, 6, 8)
males    <- c(1, 0, 1, 2, 6, 4, 8, 11, 13, 5, 12, 6, 11, 12, 14)
```

2a. Write a function `nll_mono(par, y, d)` that implements your NLL from 1d, where `par` holds a and b . Check it: `nll_mono(c(10, 1), males, dist_km)` should return about 33.84 if you used `dpois()`, or about -121.31 if you coded the formula yourself and dropped the constant term. (Why can the second number be negative?)

Your value: _____

2b. Evaluate the NLL at your starting values from 1c. Then use `optim()`, starting from those values, to find the MLE.

NLL at your starting values: _____

$\hat{a}$ = _____ $\hat{b}$ = _____ NLL at the MLE: _____ convergence code: _____

Did `optim()` print any warnings? If so, what do you think caused them, and do they mean the answer is wrong?

2c. Plot the data with your fitted curve and your hand-guessed curve from 1c. How far from the road does the expected chorus reach half its full size?

Half-size distance: _____ km

2d. A skeptic says the road has no effect: every pond has the same expected count λ , whatever its distance. What is the MLE of λ under that model, and what is its NLL? (Keep or drop the constant term the same way you did in 2a, so the two NLLs are comparable.) Which model do the data support more? Does the size of the difference convince you?

$\hat{\lambda}$ = _____ NLL: _____

2e. Path A: write out the prompt you used. Did the AI’s approach match your plan from 1e? What did you have to check or fix? **Path B:** what was hardest to translate from your plan into code?

If you finish early. A colleague prefers a different saturating curve, $\lambda(d) = \frac{ad}{b+d}$. Fit it the same way and compare its NLL with the monomolecular model’s. What do you conclude?

Reflection (~10 min)

In Act 1, a single visit detected this frog only about 40% of the time. What does that suggest about what the counts in this Act really measure? Also record any changes you made, new insights, and any “muddiest points” or topics you want to review.
